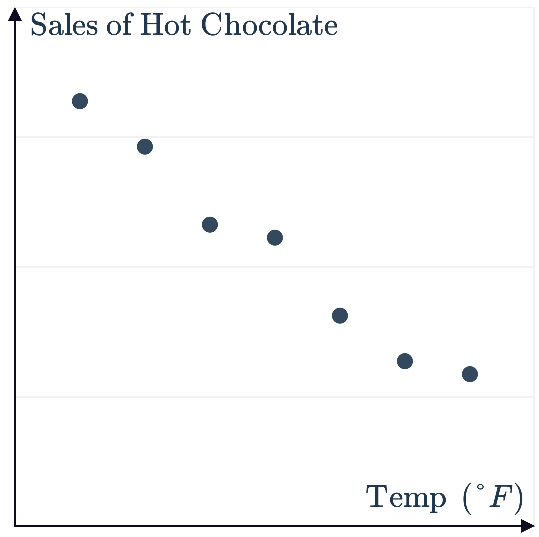
Extension: Fitting functions to quadra[redacted]ata

1 For each of the following scatter plots:

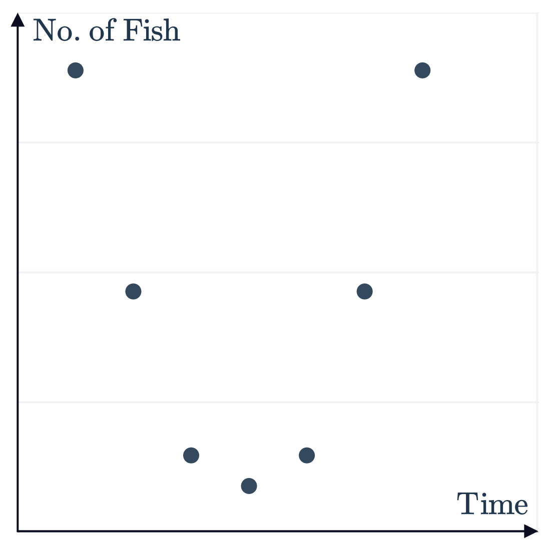
- **i** State the type of function that best models the data.
- **ii** State whether the slope of the line (for a linear model) or coefficient of x^2 (for a quadratic model) is positive or negative.

a

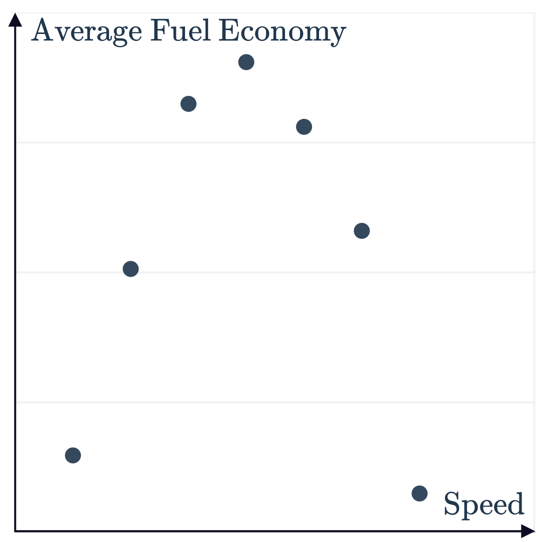
b



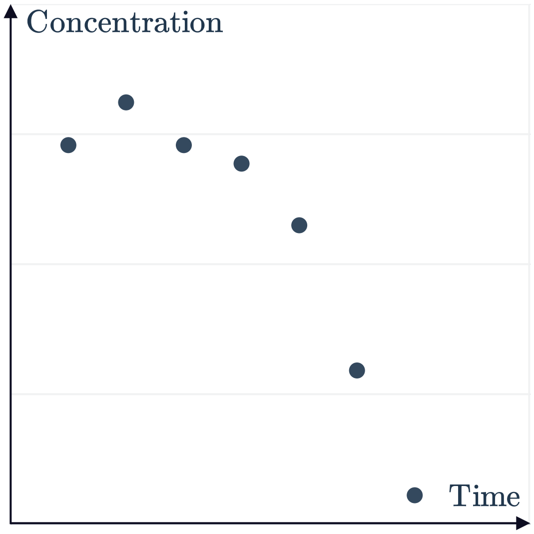
c

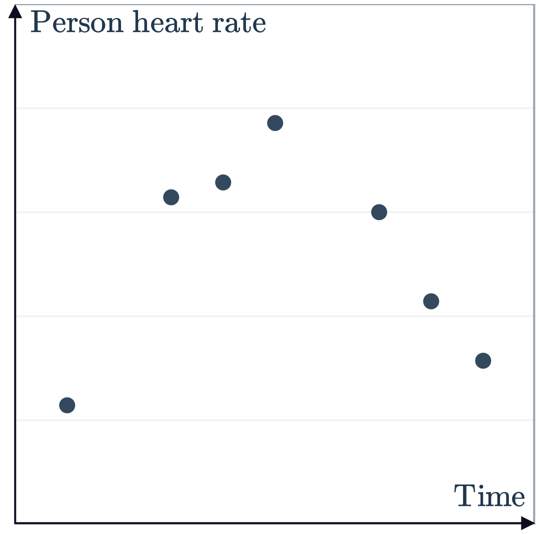


d



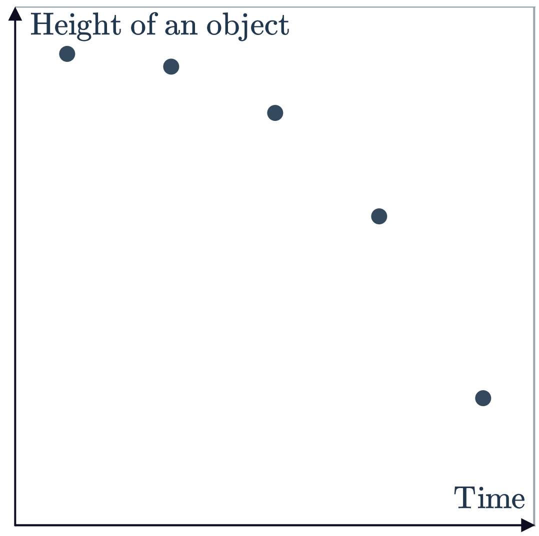
e



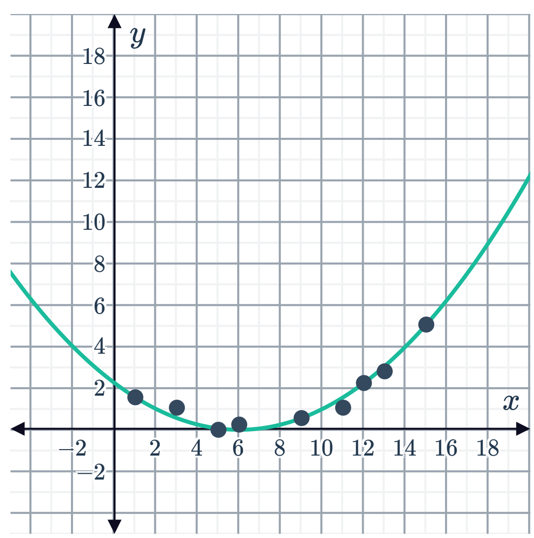


g

h



2 Nine data points have been plotted below with a quadratic curve of best fit:

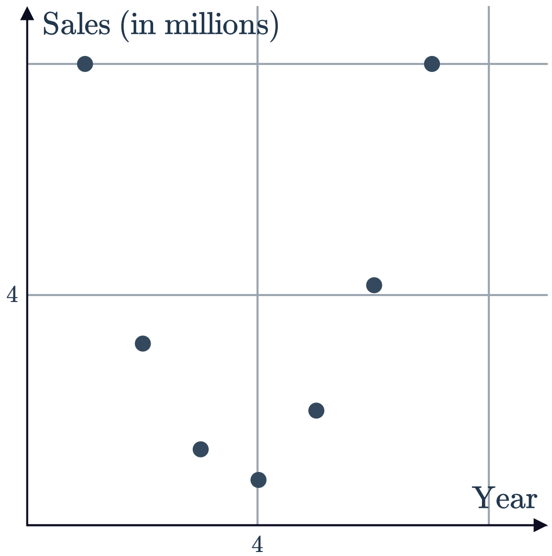


- **a** Predict the y-value of a point with an x-value of 2.
- **b** Determine whether the following points would be predicted by the quadratic curve of best fit:
  - **i** $(9, 3)$
  - **ii** $(3, 4)$
  - **iii** $(12, 0)$
  - **iv** $(14, 4)$

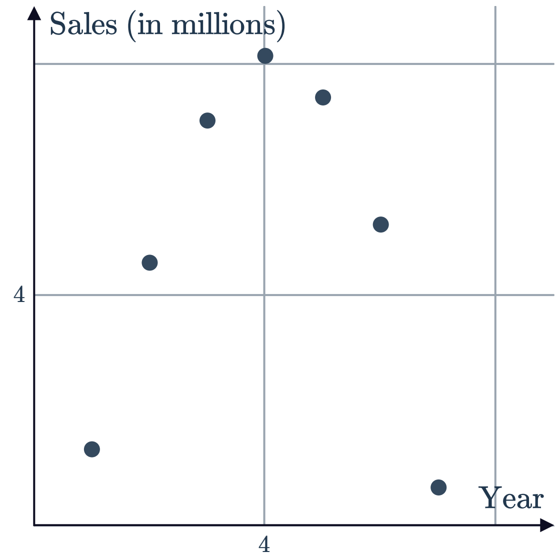
3 Which one of the following types of functio[redacted]an appropriate model for the data shown on each graph?

- Linear, $f(x) = mx + b$
- Quadratic, $f(x) = ax^2 + bx + c, a > 0$
- Quadratic, $f(x) = ax^2 + bx + c, a < 0$

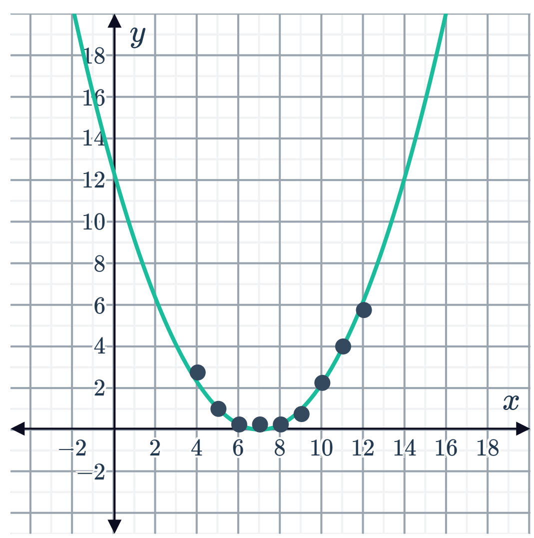
a



b



4 Nine data points have been plotted below with a quadratic curve of best fit:



a Predict the y-value of a point with an x-value of 13.

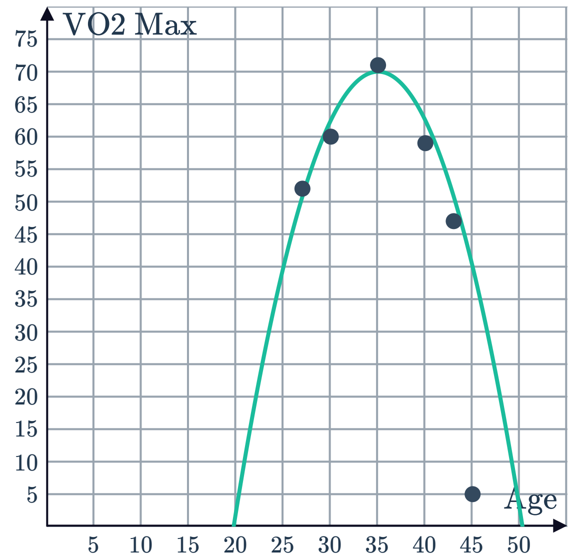
b Determine whether the following points would be predicted by the quadratic curve of best fit:

i $(3, 4)$ **ii** $(14, 10)$

iii $(2, 9)$ **iv** $(15, 15)$

Applications

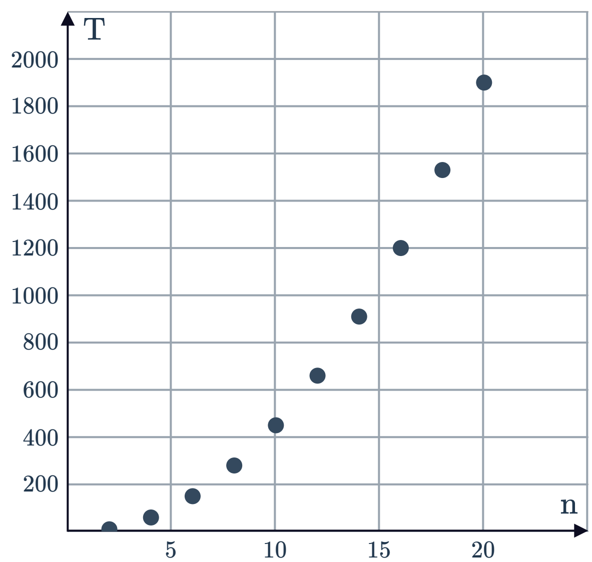
5 The plotted points in the given graph represent the age and VO2 max of 6 professional athletes, where VO2 max is a measure of aerobic fitness. The function $y = -0.3x^2 + 21x - 297.5$ has been fitted to model the relationship between age and VO2 max.



a According to the model, what will be the VO2 max of an athlete when they retire at the age of 32?

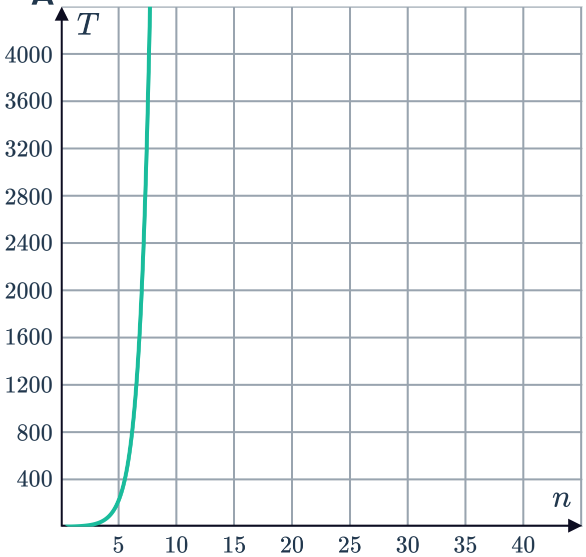
b According to the model, at what age do athletes reach their maximum VO2 max?

6 A computer program compares and orders the scores of all students who sit an exam. The time taken (T, in milliseconds) for the program to completely order all students is plotted for different numbers of students, n:

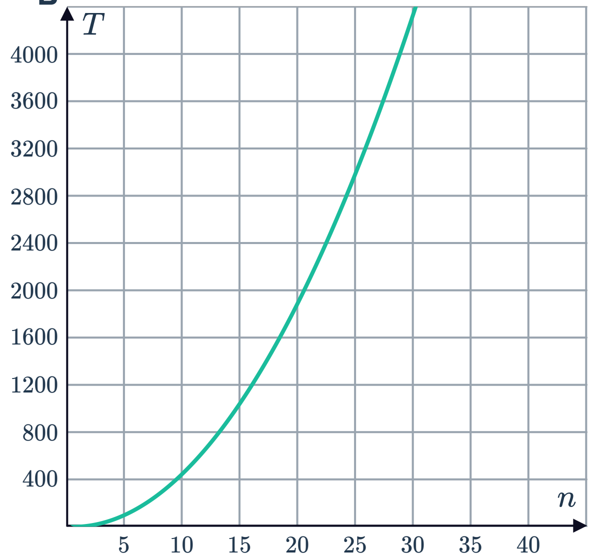


Which one of the following model will best describe the run time of the computer program as the number of students continues to increase?

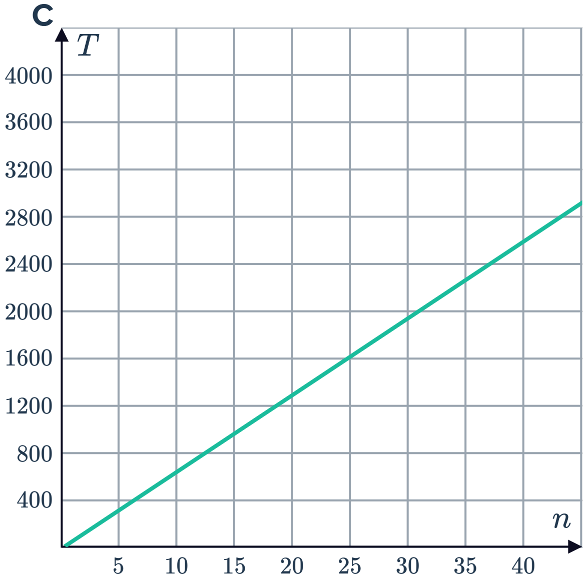
A


T
4000
3600
3200
2800
2400
2000
1600
1200
800
400
n
5
10
15
20
25
30
35
40

B


T
4000
3600
3200
2800
2400
2000
1600
1200
800
400
n
5
10
15
20
25
30
35
40

C


T
4000
3600
3200
2800
2400
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1600
1200
800
400
n
5
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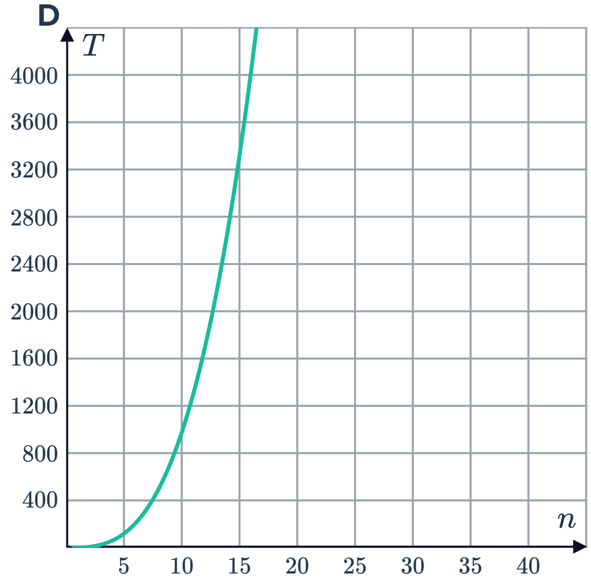
25
30
35
40

D


T
4000
3600
3200
2800
2400
2000
1600
1200
800
400
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5
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25
30
35
40

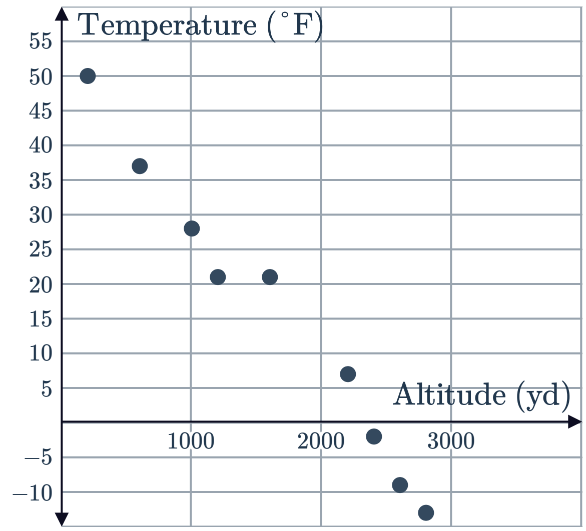
7 The scatter plot below shows the altitude a[illegible]erage annual temperature of ten randomly selected locations on Earth.

a What type of function best models the relationship between the altitude of a location and its average annual temperature?

b It was found that the function $y = -0.025x + 59$ can be used to model the relationship. Graph the function on the same set of axes as the plotted points.

c Complete the table by using the model to approximate the average annual temperature of locations at the given altitudes.

d Dylan starts a mountain hike at an altitude of 940 yd and plans to reach the summit at an elevation of 1850 yd. According to the model, by how much will the temperature decrease?



Altitude (x)	Average Annual Temperature (y)
500	
1000	
2000	

8 Hermione has just purchased a new car and wants to know how the speed at which she drives changes the gas consumption of her car. With the help of a friend, she records the gas consumption at several different speeds.

a Plot the data points from the table.

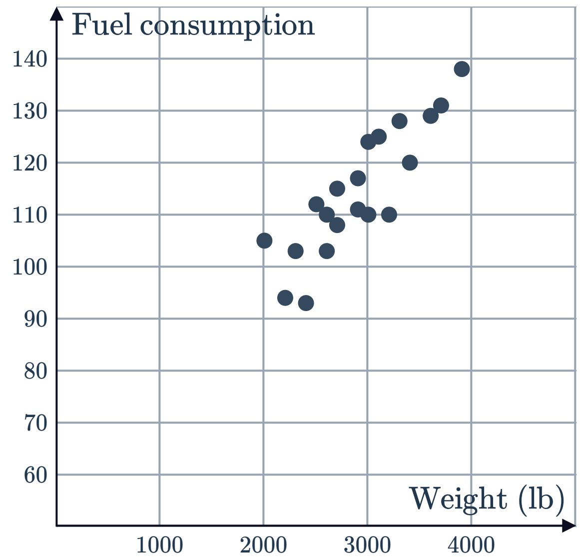
b Using the data from your plotted graph, what type of model would be most appropriate?

c A graphing utility has fitted the data from the table to a quadratic model. The calculator's output is shown below.
Use these results to build a quadratic model from the data, giving each of the constants correct to two decimal places.

Speed, x km/hr	Fuel consumption, y L/100km
30	13
40	7
50	5
60	6
70	14
80	23

QuadReg
$y = Ax^2 + Bx + C$
$A = 0.02053571$
$B = -2.053214$
$C = 56.15$

9 A sample of 20 cars were weighed and thei[redacted]age fuel consumption (measured in gallons per mile) measured.



a What type of function best models the relationship between the weight of a car and its average fuel consumption?

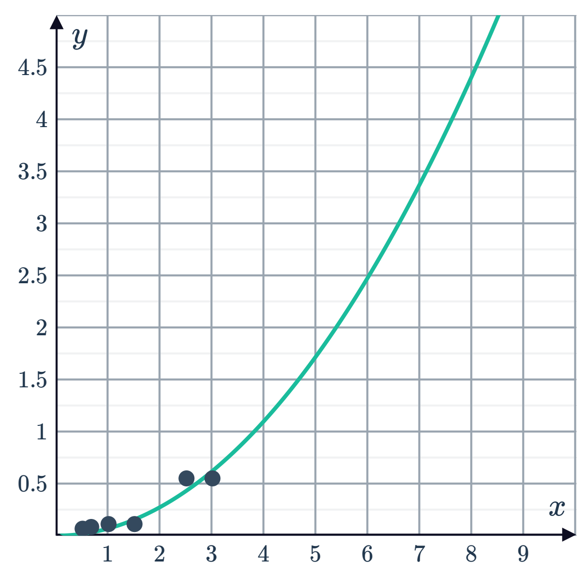
b A car manufacturer needs to provide information on the average fuel consumption of its cars. Rather than test each of them on the road, the manufacturer uses the function $y = 0.02x + 55$ to fit the model. Graph the line on the same set of axes as the plotted points.

c Complete the table by using the function $y = 0.02x + 55$ to approximate the average fuel consumption of cars with the given weights.

Weight (x)	Average Fuel Consumption (y)
3200	
3100	
1500	

d Bob lives in the city, so he wants to purchase a car that is relatively fuel efficient. Two cars that he is considering weigh 1600 lb and 2900 lb respectively. According to the model, which car should he choose if fuel consumption is the only consideration?

10 The distance d in kilometers that Emma runs was measured at different times t minutes after she started. The following quadratic curve of best fit was graphed:



a Using the curve of best fit, find the predicted distance Emma runs after:

i 2 minutes **ii** 8 minutes

b Which of the above predictions is less reliable? Explain your answer.

11 A social researcher claims that the longer p[redacted] stay in their job, the less satisfaction they gain from their work. She asked a sample of people how many years they had been employed in their current job and to rate their level of satisfaction out of 10. The results are presented in the table.

Number of years employed	Satisfaction Rating
2	9
3	5
5	4
6	2
8	5
10	7
12	8
13	7
15	9

a Create a scatter plot for the data collected.

b Does the scatter plot support the social worker's claims?

c Write the general form of the equation that would be best suited to model the relationship between the number of years employed and satisfaction with the job.

d The social researcher herself has been employed in her current job for 4 years and rates her satisfaction with her work a 10 out of 10. Find the difference between the satisfaction rating approximated by the model $y = 0.5x^2 - 6x + 20$ and her actual rating.